🗺️ October 30 Execution Pipeline (Visual Blueprint)

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Skip reading the paper text-by-text. Follow this exact functional pipeline in your Colab notebook:

```
[1. Download ShipsEar]
        |
        ▼
[2. Chunk Audio] ⟶ Slice into 5-second clips (Drop corrupt files) ⟶
Total: ~2303 samples
        |
        ▼
[3. Split Datasets]
        ├── Dataset A: Random Split (70% Train / 15% Val / 15% Test)[cite:
12]
        └── Dataset B: Chronological Split (First 70% Train, Rest Split 1:1)
[cite: 12]
        |
        ▼
[4. Feature Extraction] ⟶ Generate 2D matrices using `librosa`:
        ├── STFT  (Linear bins)[cite: 10]
        ├── FBank (Mel-scale filter banks)[cite: 10]
        └── MFCC  (Cepstral coefficients)[cite: 10]
        |
        ▼
[5. ResNet18 Baseline]
        ├── Modify input channel: change first conv layer to
`in_channels=1`[cite: 13]
        └── Train & log accuracies to reproduce Table 2[cite: 2, 13]
```

📊 Table 2 Target Reference (Your Goal)

Your code must run and match these exact baseline benchmarks from the paper:

Feature Extraction	Dataset A Accuracy Target	Dataset B Accuracy Target
STFT	76.9%	

| 60.1%

| | **FBank** | 93.8%

| 88.2%

| | **MFCC** | 93.8%

| 85.5%

|

⚡ 3-Step Action Checklist for Colab

1. **Audio Prep Script:** Write a short loop using `librosa.load()` to read files, slice them into strict 5-second windows, and save them into class folders.
2. **The Split Code:**

- For Dataset A: Use `sklearn.model_selection.train_test_split(..., random_state=42)`.
- For Dataset B: Sort filenames by timestamp/name string and slice the array index directly at 70%.

3. **ResNet Hack:**

```
import torchvision.models as models
import torch.nn as nn

model = models.resnet18(pretrained=False)
model.conv1 = nn.Conv2d(1, 64, kernel_size=7, stride=2, padding=3,
bias=False)  # Change input channels to 1
```